OS - Orange Problem

1A)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ make test_objects
./test_objects
make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

1B)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

2A)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ make test_tree
./test_tree
make: 'test_tree' is up to date.
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

2B)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ find .pes/objects -type f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/6b/05175a124d81a23519efda73626d9b4f21fda45d02c297211a2267b92af330
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/c1/566bbeaadb1ebff6cb7b82ae783d7292c9b5df43520be45e4f80968c76fe11
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/5b/9e97743b60743ce97f72138bb5c0668dc26d2e8f2d471349ac4d86b72970cb
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ xxd .pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a6865
2bc | head -20
00000000: 626c 6f62 2038 0047 6f6f 6462 7965 0a    blob 8.Goodbye.
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

3A)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ make pes
make: 'pes' is up to date.
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
Initialized empty PES repository in .pes/
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ ./pes add file1.txt file2.txt
./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  README.md
  untracked:  tree.h
  untracked:  pes.c
  untracked:  pes.h
  untracked:  commit.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  index.h
  untracked:  object.c
  untracked:  test_tree
  untracked:  index.c
  untracked:  tree.c
  untracked:  Makefile
  untracked:  .gitignore
  untracked:  commit.h
  untracked:  test_objects
```

3B)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776349313 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776349313 6 file2.txt
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

4A)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ ./pes init

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
Initialized empty PES repository in .pes/
Committed: 6b05175a124d... Initial commit
Committed: 5b9e97743b60... Add world
Committed: c1566bbeaadb... Add farewell
```

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ ./pes log
commit c1566bbeaadb1ebff6cb7b82ae783d7292c9b5df43520be45e4f80968c76fe11
Author: PES User <pes@localhost>
Date:   1776349511

    Add farewell

commit 5b9e97743b60743ce97f72138bb5c0668dc26d2e8f2d471349ac4d86b72970cb
Author: PES User <pes@localhost>
Date:   1776349511

    Add world

commit 6b05175a124d81a23519efda73626d9b4f21fda45d02c297211a2267b92af330
Author: PES User <pes@localhost>
Date:   1776349511

    Initial commit

aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

4B)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/5b/9e97743b60743ce97f72138bb5c0668dc26d2e8f2d471349ac4d86b72970cb
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6b/05175a124d81a23519efda73626d9b4f21fda45d02c297211a2267b92af330
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/c1/566bbeaadb1ebff6cb7b82ae783d7292c9b5df43520be45e4f80968c76fe11
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

4C)

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
c1566bbeaadb1ebff6cb7b82ae783d7292c9b5df43520be45e4f80968c76fe11
ref: refs/heads/main
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```

Final Implementation

```
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)
```

```
--- First Commit ---
Committed: e461e4c62296... Initial commit

Log after first commit:
commit e461e4c62296821c836c6b8718156b65c4e1758d8c32bb586c8994de6ca9a834
Author: PES User <pes@localhost>
Date:   1776349737

    Initial commit


--- Second Commit ---
Committed: 72a600ad012e... Update file.txt

--- Third Commit ---
Committed: fde8bef6cdd3... Add farewell

--- Full History ---
commit fde8bef6cdd3ddb826dbcb491ff0b8daa86df6d1b6b32b80620a441ffa542254
Author: PES User <pes@localhost>
Date:   1776349737

    Add farewell

commit 72a600ad012e9fe109fb956b1312b1a03e29f2b56c027136efbb3b491fa98277
Author: PES User <pes@localhost>
Date:   1776349737
```

```
    Update file.txt

commit e461e4c62296821c836c6b8718156b65c4e1758d8c32bb586c8994de6ca9a834
Author: PES User <pes@localhost>
Date:   1776349737

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
fde8bef6cdd3ddb826dbcb491ff0b8daa86df6d1b6b32b80620a441ffa542254
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/72/a600ad012e9fe109fb956b1312b1a03e29f2b56c027136efbb3b491fa98277
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e4/61e4c62296821c836c6b8718156b65c4e1758d8c32bb586c8994de6ca9a834
.pes/objects/fd/e8bef6cdd3ddb826dbcb491ff0b8daa86df6d1b6b32b80620a441ffa542254

=== All integration tests completed ===
aadi@serveraadi:~/PES2UG24CS005-pes-vcs$
```